Noise in Disguise: What Powers Physics-Inspired GNN Colouring

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Abstract

Physics-inspired graph neural networks (PI-GNNs) colour a graph by training a network to minimise a Potts energy, the expected number of conflicts under its soft colour assignment, and rounding the result. Which part deserves the credit: the network and its optimiser, the soft output, or the decoder and repair? We intervene on one at a time, on 68 DIMACS graphs at one colour below the chromatic number, with checks at the chromatic number and on planted random graphs. Training produces a single confident colouring (in the median graph, 99.7% of vertices end with confidence at least 0.99) that is usually not single-flip optimal (42% of runs after 10^5 steps), ties the greedy DSATUR heuristic after repair, and loses to a time-matched tabu search. The network is not what makes it good: without dropout it matches optimising one logit vector per vertex directly, with no network, and the same optimisation with Gaussian noise beats PI-GNN on 44 of 68 graphs. The soft output encodes neighbour coordination, yet a decoder given no learned input does as well, and local search gains nothing from starting at the trained colouring. Descent on the Potts energy does the work and noise decides how well; the network's advantage is noise in disguise.

1 Introduction

A k-colouring of a graph $G = (V, E)$ gives every vertex one of k colours. An edge whose endpoints share a colour is a *conflict*, and the chromatic number $\chi(G)$ is the fewest colours that allow none. Minimising conflicts is NP-hard, and unsupervised graph neural networks are a recent line of attack. The physics-inspired GNN (PI-GNN) of Schuetz et al. (2022b) trains a fresh network for each graph. Its softmax output is a *soft colouring* P, its loss is a Potts energy equal to the expected number of conflicts, and its answer is the argmax of P.

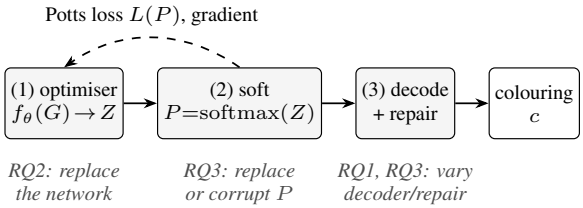


Figure 1: The three ingredients of PI-GNN colouring and where each research question intervenes. The graph enters both the network and the loss.

When this pipeline produces a good colouring, what deserves the credit? Figure 1 separates three ingredients that are normally evaluated together: (1) the network and its optimiser, (2) the soft output P, and (3) the decoder and any repair. Critiques showing that greedy algorithms match or beat PI-GNNs (Angelini and Ricci-Tersenghi, 2023; Boettcher, 2023) tell us *that* the pipeline can fall short, not *which* part is responsible. We ask: *which ingredient of PI-GNN is responsible for the quality of its colourings: the GNN optimiser, the learned soft output, or the decoding and repair?*

We intervene on one ingredient at a time (Figure 1). **RQ1**: what does training produce? Does the decoded output reach a discrete local minimum, and how good is it? **RQ2**: is the network needed? Does it beat optimising one free logit vector per vertex on the same loss? **RQ3**: does the learned P help a stronger decoder more than an uninformative or corrupted input would?

The main experiments use $k = \chi(G) - 1$ colours. Zero conflicts are then impossible, so methods are compared on a graded conflict count rather than a success rate that saturates on easy graphs; a second protocol checks $k = \chi$ and planted random graphs.

Our answer is that **descent on the Potts loss does the work and noise decides how well**. Training yields one confident colouring stuck short of a local optimum (RQ1); the network's edge over

free logits is its dropout noise (RQ2); and decoders find nothing left to decode (RQ3). The protocol itself, with its uninformative-input and search-start controls, is a second contribution.[1]

2 Background and Related Work

Learning for combinatorial optimisation. Learned methods either construct solutions or guide classical solvers (Bengio et al., 2021). For colouring, Lemos et al. (2019) trained a supervised GNN to decide k-colourability. Unsupervised approaches instead minimise a differentiable relaxation on each instance. Karalias and Loukas (2020) relax the objective to an expectation under independent vertex distributions and recover a solution by derandomised rounding (Raghavan, 1988), with a guarantee in terms of the relaxed loss. PI-GNN (Schuetz et al., 2022a,b) applies the same idea with a QUBO or Potts energy (Wu, 1982) and a GraphSAGE network (Hamilton et al., 2017), reporting results competitive with classical solvers on DIMACS graphs (Johnson and Trick, 1996).

Critiques and ablations. Angelini and Ricci-Tersenghi (2023) and Boettcher (2023) showed, end to end, that simple greedy algorithms match or beat PI-GNN on maximum independent set and max-cut. Vanderbush and Weber (2026) include a direct-logit (depth-zero) ablation of a modified PI-GNN for colouring. We make that comparison the central object and add what an ablation alone leaves open: a shared start, separate tuning, matched time, noise controls and a common repair step.

Classical references and training dynamics. DSATUR (Brélaz, 1979) colours the most constrained vertex first. TabuCol (Hertz and de Werra, 1987) is a local search that accepts worsening moves and forbids recent reversals, and underlies state-of-the-art hybrids (Galinier and Hao, 1999). Two facts about training matter for RQ2: a network moves its output through shared parameters, so one step moves many vertices at once (Jacot et al., 2018), and dropout (Srivastava et al., 2014) injects noise into training.

[1] Code and results: https://github.com/guretskiysemyon/GraphsML. Implementation, experiments and drafting used AI coding assistants extensively; all numbers come from the accompanying code and results, and the design, interpretation and final text are our own.

3 Setup

Relaxation. A soft colouring $P \in \mathbb{R}^{n\times k}$ has one probability row p_v per vertex, obtained from logits by $P = \mathrm{softmax}(Z)$. The Potts loss and the hard conflict count are

$$L(P) = \textstyle\sum_{\{u,v\}\in E} p_u^\top p_v,$$
$$C(c) = \textstyle\sum_{\{u,v\}\in E} \mathbf{1}[c_u = c_v].$$

Since $p_u^\top p_v$ is the probability that u and v draw the same colour, $L(P)$ is the expected conflict count when each vertex samples independently, and $L = C$ when every row is one-hot.

Local optimality and repair. Let $M_v(a)$ be the number of neighbours of v with colour a. Recolouring v can remove at most $\gamma_v = M_v(c_v) - \min_a M_v(a)$ conflicts, and a colouring is *single-flip optimal* (SFO) when $\gamma_v = 0$ for all v. Our shared repair, *strict descent*, repeatedly applies a move with the largest positive γ_v and stops at an SFO colouring. We report raw conflicts C (how good a method's own answer is) and C_{desc} after descent (how good a *starting point* it is); in RQ2 the two disagree.

Lemma 1 (cf. Rosenberg, 1972)**.** *If P^* is a strict local minimum of L over all row-stochastic matrices, every row is one-hot and every single-vertex recolouring strictly increases C.*

The proof (Appendix B) rests on L being affine in each row: with other rows fixed, $L = B + p_v^\top q_v$ with $q_v = \sum_{u\in N(v)} p_u$. Training, however, moves logits rather than probabilities, and for the Potts loss

$$\frac{\partial L}{\partial z_{v,a}} = p_v(a)\big[q_v(a) - p_v^\top q_v\big]. \qquad (1)$$

Suppose p_v is nearly one-hot on colour c. For $a \neq c$ the factor $p_v(a) \approx e^{-\Delta}$ is tiny, where Δ is the gap between the top two logits; for $a = c$ the bracket is $\approx q_v(c) - q_v(c) = 0$. So every logit gradient of a *saturated* vertex vanishes, even when another colour would remove many conflicts.

PI-GNN and benchmark. We reimplement PI-GNN in PyTorch: trainable Gaussian vertex embeddings (dimension 85), two mean-aggregation GraphSAGE layers (hidden width 30, ReLU, dropout 0.47), trained full-batch with AdamW (Loshchilov and Hutter, 2019). The benchmark is the 68 DIMACS graphs whose chromatic number

is proven (Held et al., 2012; Gualandi and Malucelli, 2012), with 11 to 1,000 vertices from 14 families: 5 seeds per graph, 10^5 steps, no early stopping, dropout off at evaluation. Every model type is tuned separately by the same two-stage random search on 13 disjoint synthetic graphs, then frozen.

Second protocol. Independent runs at $k = \chi$ check the same questions on the 12 DIMACS graphs of Schuetz et al. (2022b), 12 harder ones, and 27 planted 5-colourable random graphs near the clustering threshold (Zdeborová and Krzakala, 2007), pairing runs by graph and seed; ties are exact.

Comparison rule. The graph is the unit of analysis: we average the 5 seeds per graph and compare paired means X_g, Y_g through $d_g = (X_g - Y_g)/\max(Y_g, 1)$. A graph is a practical tie if $|X_g - Y_g| < 1$ conflict or $|d_g| \leq 5\%$, otherwise a win or a loss. We report wins/ties/losses (W/T/L), the median d_g and a two-sided Wilcoxon signed-rank test (Wilcoxon, 1945). A directional claim needs a small p (Bonferroni-corrected over the primary comparisons), a median beyond 5% and more wins than losses; anything else is "not distinguishable", which is not a claim of equality. The rule was fixed before the main RQ2 results were computed; it rests on medians because conflict counts span four orders of magnitude.

4 Experiments

4.1 RQ1: Sure but stuck

What does training produce? One confident colouring, usually short of a local optimum. Figure 2 tracks the argmax output over training. The fraction of SFO runs rises from 0% to 42% $(143/340)$, and the median share of conflicts that descent can still remove falls from 95% to 10%. Both curves flatten after $5{\cdot}10^4$ steps. Extending 13 still-improving graphs to $3{\cdot}10^5$ steps lowers conflicts after descent on 12, yet SFO runs go from $0/65$ to only $1/65$. Longer training finds better basins without reaching their bottoms: it delivers the one-hot rows of Lemma 1, not its local optimality.

The remaining errors sit on a handful of vertices (median 2.5 improvable vertices per run, under 1% of the graph), and those vertices are *confident*. In the median graph, 99.7% of vertices have $\max_a p_v(a) \geq 0.99$, and so do 100% of the improvable ones; their median top-two logit gap

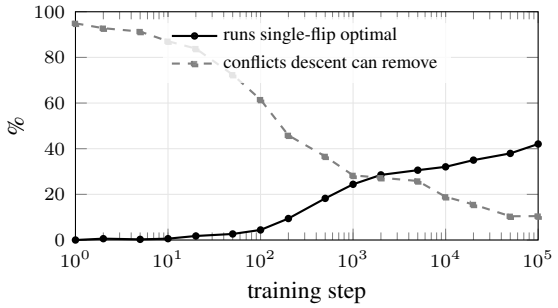


Figure 2: RQ1: PI-GNN's argmax output over training (68 graphs, 5 seeds). Solid: share of runs single-flip optimal. Dashed: median relative conflict reduction strict descent still achieves.

PI-GNN + descent vs.	W/T/L	median d
random colouring + descent	62/5/1	-38.7%
own epoch-1 output + descent	59/8/1	-32.5%
DSATUR + descent	16/28/24	0.0%
TabuCol, time-matched	14/13/41	–

Table 1: RQ1: quality of the trained endpoint at 10^5 steps, counted from PI-GNN's side (negative d favours PI-GNN). Mean conflicts: PI-GNN 114.9, TabuCol 70.5.

is about 150, so the better colour has probability $\approx e^{-150}$. The "distribution" is a single colouring. By Eq. (1), a confident wrong vertex gets almost no gradient towards the better colour: one flip away, yet out of reach. This is a plausible mechanism, not a proven one, since shared parameters can still move such a vertex indirectly.

Is the endpoint at least good? Table 1 compares PI-GNN after descent with references given the same descent. Training clearly beats random and epoch-1 starts, but it is not distinguishable from DSATUR, a 1979 greedy heuristic, and a time-matched TabuCol finds fewer conflicts on 41 graphs. The queen family makes the point sharply: 85% of queen runs are already SFO, yet TabuCol wins on all 12 queen graphs. A local minimum can be shallow.

4.2 RQ2: Noise in disguise

Is the network needed? Only as a source of noise. The graph already enters the Potts loss, so a network is not needed to "use" it. The natural network-free competitor, arm B, optimises the logit matrix Z directly from PI-GNN's exact initial output, so only the optimisation differs. The other arms each change one thing (Table 2); two need a word. B-noise adds Gaussian noise of a

preset scale σ to Z during training, reusing B's tuning. D is PI-GNN linearised at initialisation, $Z = Z_0 + J_0\delta$ with a frozen Jacobian, which tests whether the network acts as a fixed graph-dependent preconditioner.

The verdict flips with repair. Free logits give *better raw* colourings than PI-GNN (32 wins, 13 losses, $p = 0.002$). After descent, PI-GNN is better (26 wins, 17 losses, median 6%, $p < 0.001$), also at matched wall-clock time, where B gets about $2.9\times$ more steps. The SFO column explains why: B ends single-flip optimal in all 340 runs, so descent adds nothing to it, while PI-GNN stops short in 58% of runs but in slightly deeper basins. By the *mean*, B is still lower (108.6 vs. 114.9), as PI-GNN's wins come on low-conflict graphs.

The advantage is noise, not architecture. Remove dropout and PI-GNN's edge is gone: A-no-drop loses to PI-GNN on 21 graphs (7 wins) and is not distinguishable from B after descent (and in a second-protocol pilot, dropout won all 8 matched learning-rate and width pairs). Add noise to free logits and the edge is overturned: after descent they beat PI-GNN on 35 graphs at $\sigma = 1$ and on 44 at $\sigma = 2$, losing on 3, and at $\sigma = 2$ beat plain B on 53 graphs with no losses. Mean conflicts (Tables 1–2) read as a ladder: network 114.9, free logits 108.6, with noise 92.5 and 83.4, time-matched TabuCol 70.5.

Dropout and logit noise reshape confidence differently. Dropout delays PI-GNN's commitment (69% of vertices confident at step 100 vs. 89% without it) but ends far more saturated (median logit gap 204 vs. 18). Logit noise at $\sigma = 2$ commits earlier (82% vs. 55% at step 1k) yet stops sharpening (93% confident vs. 100%, gap 8 vs. 18 for plain B). So the best arm keeps some uncertainty late, as a temperature would (Kirkpatrick et al., 1983; Peterson and Söderberg, 1989), while dropout pushes outputs into the saturated regime of Eq. 1; this link to quality is correlational, not causal.

What does the network contribute? Aggregation helps only within the neural recipe: the node MLP C ties PI-GNN on raw colourings (20/28/20) but loses after descent (5/35/28). D, with its Jacobian frozen at initialisation, never beats PI-GNN after descent (0/10/58), and B ties B-rand (14/42/12): the initial output gives no head start.

4.3 RQ3: Nothing left to decode

Is it the soft output or the decoder? Neither adds much to the colouring that P already encodes. We fix the decoder and vary its input. The decoder is derandomised conditional expectation (D-CE) (Raghavan, 1988; Karalias and Loukas, 2020): it fixes vertices in DSATUR order, giving v the colour a that minimises the number of fixed neighbours coloured a plus $\sum p_u(a)$ over unfixed neighbours u. This guarantees $C \leq L(P)$, but on a one-hot P, where $L = C$, it only promises to match argmax. The first term reads the graph, so D-CE needs no learned information: on uniform P it is DSATUR, and on near-uniform P it beats raw argmax on all 68 graphs (median -93%). This near-uniform input is our *graph-only* baseline. We also corrupt the trained P in three ways: the untrained output, rows shuffled across vertices, and colours permuted independently within each row (same confidence, no neighbour agreement).

Real information, but no more than the graph alone. Trained P beats all three corruptions (Table 3; medians -22 to -30%, at most 3 losses), including colour-scrambled P with identical confidence, so its useful content is *coordination* between neighbours, not confidence. Against the graph-only baseline, however, trained P is not distinguishable (27/28/13, median 0%, $p = 0.23$), echoing RQ1's tie with DSATUR. Scrambled P is worse than no P at all (near-uniform wins 41/24/3), which is why beating a corruption is a weak test. The tie hides a family split (see Limitations).

Most of the decoder's gain is repair. On trained P, D-CE beats raw argmax (33/35/0, median -11.4%). Once argmax also gets descent, the gain shrinks to -2.8% (22/46/0): inside the tie band, though never worse and cheaper overall (21 vs. 30 s). Best-of-32 sampling adds nothing (2/65/1) at $26\times$ the cost, presumably because saturated rows make samples largely repeat argmax. The second protocol agrees (Table 3, bottom), and TabuCol does as well from a random start as from the trained colouring: whatever P knows, local search rebuilds within 10^4 iterations.

The optimiser still decides. Decoding all nine RQ2 arms with D-CE narrows their spread (median within-graph range -10%) but keeps their order: B-noise-2 still beats PI-GNN (35/29/4), D stays last, and only B-noise-2's P beats the graph-

Arm	change relative to A	vs. A, W/T/L (arm better / tie / A better) raw C	after descent	median d (desc.)	mean C_{desc}	SFO	logit gap
A	PI-GNN (reference)	–	–	–	114.9	42%	204
B	free logits from A's start	32/23/13	17/25/26	+6.0%	108.6	100%	18
B-rand	as B, random start	28/21/19	20/20/28	+10.8%	107.1	100%	19
A-no-drop	dropout removed	10/9/49	7/40/21	+8.7%	115.7	17%	18
B-noise-0.5	B + logit noise, σ=0.5	33/21/14	20/19/29	+5.8%	109.2	100%	19
B-noise-1	B + logit noise, σ=1	46/20/2	35/30/3	−15.9%	92.5	100%	13
B-noise-2	B + logit noise, σ=2	53/13/2	44/21/3	−27.5%	83.4	100%	8
C	node MLP: aggregation removed	20/28/20	5/35/28	+7.8%	120.4	41%	139
D	linearised at init, frozen J_0	14/12/42	0/10/58	+36.6%	130.4	42%	23

Table 2: RQ2 at 10^5 steps (68 graphs, 5 seeds). W/T/L and median $d = (\text{arm} - \text{A})/\max(\text{A}, 1)$ are graph-level; negative d favours the arm. SFO: share of runs single-flip optimal. Logit gap: median top-two logit difference (saturation). B vs. A is the primary comparison.

first vs. second	W/T/L	median d
trained vs. colour-scrambled	48/20/0	−25.5%
trained vs. vertex-shuffled	56/12/0	−29.9%
trained vs. untrained	51/14/3	−22.2%
trained vs. near-uniform	27/28/13	0.0%
D-CE vs. argmax, raw	33/35/0	−11.4%
D-CE vs. argmax, after descent	22/46/0	−2.8%
Second protocol (runs paired)		
trained vs. DSATUR best of 11	15/27/12	–
D-CE vs. argmax, repaired, planted	2/22/3	–
TabuCol: trained vs. random start	22/22/22	–
same, planted graphs	13/0/14	–

Table 3: RQ3 controls. Top: D-CE + descent on frozen 10^5-step outputs; "trained" is PI-GNN's P unless stated. Bottom: one D-CE decode at $k = \chi$ (also $\chi + 1$ for TabuCol on the harder graphs), unrepaired unless marked.

only baseline (36/27/5). A better optimiser gives a better P, and decoding keeps that edge.

5 Discussion and Conclusion

Returning to Figure 1: the optimiser earns the credit through its noise, not its architecture; the soft output carries real coordination, but no more than the graph alone; and the decoder's gain is mostly repair. Descent on the Potts loss does the work, noise decides how well, and the best gradient-based procedure we found has no network at all. This sharpens the end-to-end critiques (Angelini and Ricci-Tersenghi, 2023; Boettcher, 2023): PI-GNN learns coordinated structure but commits and saturates before local optimality (Eq. 1), and what helps it, noise, is cheaper without a network.

For evaluating such methods we suggest, following fair-comparison work (Errica et al., 2020; Bechler-Speicher et al., 2025): report raw *and* repaired quality, and whether the output is still a distribution; start a local search from it and from a random colouring; use free logits as the network-free baseline; test learned outputs against an uninformative input under the same decoder; and report per-family results.

Confidence, local optimality and quality are three different things, and conflating them overstates what the network contributes. Natural next steps are tuned noisy logit optimisation against strong local search and, where instances hide global structure, convex relaxations (Appendix A).

Limitations

One architecture, one main regime ($k = \chi - 1$) and 68 heterogeneous graphs limit generality, and behaviour differs by family: the RQ3 tie hides trained P winning on all 12 queen graphs but losing on 5/12 Leighton, 2/3 DSJR, 3/5 miles and 3/6 fpsol/inithx graphs, and decoders do matter where training collapses (one queen13_13 run: 3,328 to 72 conflicts under D-CE). The second protocol uses published settings and run-level statistics, so across protocols we compare directions, not magnitudes; on its planted graphs the repaired trained colouring beats DSATUR's best of 11 (22/2/3), but one second of TabuCol reaches 13.4 conflicts. The three noise scales were fixed in advance, not tuned; noisy free logits were not compared with TabuCol; A-no-drop reuses A's settings; and logit noise and dropout differ, so reading dropout as noise is supported but not isolated. The annealed-penalty schedule we tried (Ichikawa, 2024) did not help, inference-time noise (Colantonio et al., 2024) is untested, and Appendix A rests on 12 graphs and 2 seeds. p-values are descriptive; bootstrap intervals are in the supplementary appendix.

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A Further effort: domain knowledge from a convex relaxation

Motivation. The Potts loss has no relaxation gap: its continuous minimum equals the discrete one (Rosenberg, 1972), so descending it adds no information beyond what local moves can find. Colouring has a classical relaxation that does carry global information, vector colouring by semidefinite programming (Karger et al., 1998). We use the Frieze–Jerrum form at k colours (Frieze and Jerrum, 1997): unit vectors v_i whose pairwise inner products are at least $-1/(k-1)$, the value between the vertices of a regular simplex, maximising $\sum_{\{u,v\}\in E}(1 - v_u^\top v_v)$. A proper k-colouring placed on the simplex puts every edge exactly at this floor. The problem is convex; we solve it approximately with a rank-32 Burer–Monteiro factorisation (Burer and Monteiro, 2003), in 11–132 s per graph on one CPU.

Two uses. (i) As domain-informed input to PI-GNN: the vectors initialise its vertex embeddings (*FJ init*). (ii) Alone: k-means on the vectors gives a colouring, which can also seed TabuCol. We ran both on the 12 harder DIMACS graphs of the second protocol (Section 3; 2 seeds), against PI-GNN with the best setting of a small pilot, DSATUR and TabuCol (Table A.1).

graph (k)	PI-GNN	+FJ init	SDP alone	DSATUR	TabuCol
DSJC125.5 (17)	19	10	31	21	0
DSJC250.5 (28)	55.5	42	50	60	0.6
flat300_28_0 (28)	110	96	153	124	32.2
le450_15a (15)	31	36.5	256	7	0
le450_15b (15)	35	32.5	238	7	0
le450_15c (15)	223	**0**	2	248	16.2
le450_15d (15)	215.5	**0.5**	8	250	26.8
queen10_10 (11)	10	9.5	54	10	0
queen12_12 (12)	31	26	61	29	9.6
queen14_14 (14)	32.5	32	91	32	13.4
queen15_15 (15)	41.5	36	94	42	15.2
queen16_16 (16)	43.5	46	117	44	17

Table A.1: Conflicts at k colours (second protocol, mean of 2 seeds). PI-GNN and +FJ init: best colouring found during training; SDP alone: k-means on the FJ vectors, best of 5; DSATUR: best of 11 randomised runs; TabuCol: 10 s, mean of 5 runs.

Where structure is global, geometry wins. FJ init improves PI-GNN on 10 of the 12 graphs. On the Leighton graphs le450_15c/d, built around a planted 15-colouring (Leighton, 1979), it is decisive: 0 and 0.5 conflicts, where plain PI-GNN keeps over 200 and TabuCol 16–27 after 10 s (7–24 after 60 s). The control shows why. 95–97% of edges sit exactly at the floor, so the SDP has recovered the planted colouring, and TabuCol started from its k-means colouring finds a proper colouring within 1–72 iterations (10 of 10 starts). The network is not needed: the geometry carries the solution.

Elsewhere it does not. On random, queen and the sparser Leighton graphs (le450_15a/b), the vectors alone are worse than DSATUR on 9 of 10 graphs (e.g. 54 vs. 10 conflicts on queen10_10), FJ init helps PI-GNN unevenly (most on DSJC125.5, from 19 to 10), and TabuCol stays far ahead. Earlier small-scale tests agree. In an RLF-style colour-class extraction on 13 small graphs, SDP vectors lowered the mean excess over the best known colour count from 0.615 (RLF) to 0.385, but random unit vectors in the same procedure reached 0.462 (3/35/1 SDP vs. random). As PI-GNN input features on 9 small graphs, FJ vectors beat the same rows shuffled (21/3/3) but not identity features significantly (16/6/5, Holm-corrected $p = 0.18$).

Insight. Domain knowledge from a convex relaxation helps exactly where an instance hides global structure. There it solves what neither PI-GNN nor TabuCol can, and the Potts loss, being exact, cannot supply it. Where hardness lies in a rugged landscape instead, only search helps. Testing all twelve Leighton graphs and planted-partition models would locate the boundary.

B Proof sketch of Lemma 1

With other rows fixed, $L = B + p_v^\top q_v$ where $q_v = \sum_{u \in N(v)} p_u$, so L is affine in p_v. Moving mass ϵ between two colours a, b in the support of p_v changes L by $\pm\epsilon(q_v(a) - q_v(b))$, so one of the two moves does not increase L, contradicting strictness. At a one-hot row, moving mass t towards colour a changes L by $t\,[M_v(a) - M_v(c_v)]$, which must be positive.